How Many Agents to Avoid Collapse? A Minimum Viable Population Study of Self-Training LLM Ecosystems

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Abstract

A growing fraction of web text is generated by large language models (LLMs). If future foundation models train on this text, single-model recursive training is known to cause distributional collapse. We ask the ecosystem-level analogue: is there a *minimum viable population* (MVP) of LLMs that, sharing a common synthetic-text substrate, avoids collapse? And what kind of inter-agent differentiation actually counts? We run two complementary harnesses on the same metric suite. In a fine-tuning ecosystem (DISTILGPT2 copies retraining on each others' WIKITEXT-2 outputs) even $N{=}16$ agents collapse, with perplexity rising $2.10\times$ over six iterations and a slope of 16 ppl/iter that is statistically significant ($p < 10^{-6}$); the rate of collapse falls $\sim 5\times$ from $N{=}1$ to $N{=}16$ but never reaches zero. In a retrieval-augmented ecosystem (frozen API LLMs sharing a growing post pool), $N{=}2$ *LLMs from different pretraining families* preserve full inter-agent diversity over $T{=}12$ iterations, with both pairwise distance and the Vendi score yielding regression slopes indistinguishable from zero ($p \approx 0.4$). Holding $N{=}4$ fixed and varying the axis of differentiation, four agents from different pretrained families preserve $\mathbf{1.9}\times$ the inter-agent semantic distance of four agents varied by prompt persona, by retrieval slice, or by random seed alone. Together these results sharpen the population view of model collapse: the training-time link is what matters, and the only individuality axis that buys real collapse-resistance is diversity in the prior, not the conditioning.

1 Introduction

A growing fraction of the public web is generated by large language models, and the next generation of foundation models will train on a substrate that is partly their own ancestors' output. When a single generative model is trained on its own outputs in a closed loop, the distribution it represents narrows: tails disappear first, modes collapse, and the model converges to a low-variance regime [Shumailov et al., 2024, Alemohammad et al., 2023, Dohmatob et al., 2024]. This phenomenon, *model collapse*, has been replicated across modalities, architectures, and training regimes.

Why this matters. Almost every published collapse study fixes the unit of analysis at a *single* self-training model. The web, however, is populated by *many* LLMs from different vendors, with different pretraining priors, system prompts, and retrieval pipelines. The policy-relevant question is not "does one model collapse on its own outputs" — the answer to that is yes — but rather: how many distinct LLM agents, sharing a common substrate, are enough to keep that substrate *healthy*? Borrowing the term from population ecology, is there a *minimum viable population* (MVP) [Shaffer, 1981] of LLMs below which collapse is unavoidable? And what kind of differentiation between agents counts as "different individuals"?

Gap. Three recent ecosystem-level studies each leave a piece of the question open. Hodel and West [2026] test only $M \in \{1, 2, 4, 16\}$ copies of one base model differentiated by training-data

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segment, find that the *optimal* M grows monotonically with iteration count, but never report an MVP at any fixed horizon. Wang et al. [2025] run a network of $N{=}3$ pretrained-different LLMs sharing a retrieval pool but never sweep N. Vu et al. [2025] treat $N{=}2$ model families theoretically but do not test $N{>}2$. None of these works compare multiple axes of inter-agent differentiation, and none compare the fine-tuning ecosystem to the cheaper retrieval-augmented ecosystem on identical metrics.

Our approach. We close all three gaps in one experimental program. We run two complementary harnesses on the *same* diversity metric suite (perplexity, distinct bigrams, mean pairwise embedding distance, and the Vendi-score Hill–Shannon Diversity, hereafter HSD). E1 is a fine-tuning ecosystem where N copies of DISTILGPT2 retrain on each others' WIKITEXT-2 outputs in replace mode. E2 is a retrieval-augmented ecosystem where N frozen API LLMs each query a shared, growing "internet" of their own posts. E3 fixes $N{=}4$ in the retrieval harness and varies four axes of individuality: random seed, retrieval data slice, system-prompt persona, and pretrained model family.

Quantitative preview. We find that $N{=}2$ *frozen API LLMs from different pretraining families* preserve full diversity for $T{=}12$ iterations (slope of mean pairwise distance: -0.002 ± 0.004, $p = 0.44$), while in the fine-tuning harness even $N{=}16$ agents collapse by a factor of $2.10\times$ in perplexity over only $T{=}6$ iterations. Across the four diversity axes at $N{=}4$, model-family diversity preserves $\mathbf{1.9}\times$ the terminal inter-agent distance of the next-best axis (persona or data slice). The largest single N-doubling benefit in the fine-tuning ecosystem is $N{=}8 \rightarrow N{=}16$, which cuts terminal perplexity by 26%.

Contributions.

- We introduce a population-ecology framing of recursive LLM training, formalising the MVP question and giving the first explicit MVP estimates for both fine-tuning and retrieval-based ecosystems on a common metric suite.
- We extend the only published N-vs-collapse sweep [Hodel and West, 2026] from $N \leq 16$ to a controlled statistical comparison across $N \in \{1, 2, 4, 8, 16\}$ in the fine-tuning regime and $N \in \{1, 2, 3, 5, 8\}$ in the retrieval regime, holding compute roughly constant.
- We run the first head-to-head comparison of four orthogonal axes of inter-agent differentiation (seed, data slice, prompt persona, model family) at fixed $N{=}4$, and show that *architecture diversity is the only axis that buys collapse-resistance*.
- We reconcile the divergent answers from the two harnesses into a single governance-relevant claim: it is the *training-time* link, not the inference-time loop, that drives ecosystem collapse.

Paper organization. section 2 positions the work against the single-model and ecosystem-level collapse literature. section 3 details the two harnesses, metrics, and statistical analysis. section 4 reports E1, E2, and E3 results, with MVP estimates and axis ranking. section 5 discusses limitations and what the cross-method gap means for AI deployment. section 6 closes with bottom-line answers and recommended follow-ups.

2 Related Work

Single-model collapse. Shumailov et al. [2024] formalised model collapse, showing that a single OPT-125m fine-tuned on its own WIKITEXT-2 outputs has perplexity that rises monotonically across generations and a generation distribution whose tails progressively disappear. Alemohammad et al. [2023] extended the result to image generative models under the name *Model Autophagy Disorder* (MAD), and Briesch et al. [2023] disentangled quality decay (preserved on verifiable tasks) from diversity decay (not preserved). Guo et al. [2023] introduced lexical, syntactic, and semantic diversity metrics for the recursive-training setting; we reuse the distinct-bigram metric from that work. Theoretical accounts include Dohmatob et al. [2024], who frame collapse as a change of scaling laws, and Seddik et al. [2024], who derive a maximal admissible synthetic ratio. Our N=1 condition is a direct reproduction of the Shumailov et al. [2024] "replace" regime.

Mitigation by data dynamics. Gerstgrasser et al. [2024] show that *accumulating* synthetic and real data (rather than replacing) yields a finite, bounded test error independent of iteration count, both empirically on TinyStories and theoretically. Dey and Donoho [2024] sharpen this to a universal $\pi^2/6$ asymptote. These results establish data-side mitigations; we deliberately ablate them by run-

ning our fine-tuning harness in the harshest replace regime so that diversity alone is the variable being tested.

Ecosystem-level collapse. The work most directly comparable to ours is Hodel and West [2026]: M copies of one base model, each fine-tuned on a non-overlapping segment of WIKITEXT-2, then jointly generating, pooling, shuffling, and re-splitting the synthetic data each iteration. They sweep $M \in \{1, 2, 4, 16\}$ and find the *optimal* M grows monotonically with iteration count. Our E1 reproduces their compute-controlled setup but (a) extends the sweep to $N{=}8$ at finer spacing, (b) reports an explicit MVP estimate at fixed horizon $T{=}6$, and (c) provides a direct comparison to the much cheaper retrieval-based ecosystem.

Wang et al. [2025] build a parameter-frozen retrieval ecosystem of $N{=}3$ pretrained-different LLMs sharing a growing post pool, and show that the Frobenius norm of the inter-agent embedding-distance matrix converges to a small constant. Our E2 generalises their setup along the missing N axis. Vu et al. [2025] provide a theoretical framework for $N{=}2$ heterogeneous models, validate it on Llama and Phi family LLMs, and document the substantial overlap of modern LLM pretraining corpora — empirical motivation for the ecosystem framing.

Kovač et al. [2025] treat the data side as the diversity knob, rotating across four small base models per generation and studying how training-data properties (lexical, semantic, quality) modulate distribution shift. Their result that lexical diversity *amplifies* shift while semantic diversity *mitigates* it is consistent with our finding that prompt-induced "surface" diversity buys little collapse-resistance, while pretraining-prior diversity buys a lot.

Population ecology and minimum viable population. Shaffer [1981] introduced the concept of MVP as the smallest isolated population with a high probability of long-run persistence in the wild. Traill et al. [2007] report a generic vertebrate MVP around several thousand individuals; our analogue is specifically about the *number of distinct LLMs* required to keep their shared synthetic-text substrate from converging. The Hill–Shannon Diversity / Vendi score [Friedman and Dieng, 2023] we use as a primary metric is itself imported from ecology.

Positioning. We are not the first to observe that ecosystem-level diversity helps; Hodel and West [2026] and Wang et al. [2025] both make this qualitative point. Our contribution is to (i) give the first explicit MVP estimates that pin down a number, (ii) compare four axes of "individuality" on identical metrics, and (iii) cross-check between the two methodologically very different ecosystem operationalisations that until now had not been benchmarked side by side.

3 Methodology

3.1 Problem formulation

Let $\{A_1, \ldots, A_N\}$ be a population of N generative agents sharing a common text substrate $\mathcal{S}_t$ at iteration t. At each step, every agent A_i produces a new text slice $\mathcal{S}_t^{(i)}$ and the pool is updated $\mathcal{S}_{t+1} \leftarrow \mathrm{update}(\mathcal{S}_t, \{\mathcal{S}_t^{(i)}\}_i)$. We say the ecosystem *collapses on metric* M over horizon T if the sequence $M(\mathcal{S}_1), M(\mathcal{S}_2), \ldots, M(\mathcal{S}_T)$ has a regression slope significantly different from zero in the worsening direction. The *minimum viable population* on M at horizon T, $\mathrm{MVP}_M(T)$, is the smallest N for which the slope's null hypothesis of "no decay" cannot be rejected at $p > 0.05$, or, for lower-better metrics, the smallest N for which the terminal value remains within $\mathrm{horizon_frac} \cdot (M(\mathcal{S}_T)|_{N=1} - M(\mathcal{S}_0)|_{N=1})$ of the initial value (we use $\mathrm{horizon_frac} = 0.5$).

3.2 Two complementary harnesses

Because the ecosystem question has been operationalised in two very different ways in the literature — fine-tuning ecosystems [Hodel and West, 2026, Shumailov et al., 2024] and retrieval ecosystems [Wang et al., 2025] — we run *both* on the *same* metric suite, so that MVP and axis conclusions can be cross-validated.

3.2.1 E1: Fine-tuning ecosystem

Backbone. DISTILGPT2 (82M parameters), one fresh copy per agent. **Data.** WIKITEXT-2-raw-v1 (train split as initial real data; test split, fixed across generations, for perplexity). **Per-ecosystem**

Table 1: Evaluation metrics and their interpretation.

Metric	Definition	Direction
perplexity	$\exp(\mathrm{NLL})$ on WIKITEXT-2 test	lower better
distinct-2	unique bigrams / total bigrams	higher better
mean pairwise dist.	mean off-diag. of $1 - \cos(e_i, e_j)$	higher better
Frobenius (Wang et al. 2025)	$\lVert 1 - \cos(e_i, e_j) \rVert_F$	higher better
HSD (Vendi)	$\exp(-\sum(\lambda/n)\log(\lambda/n))$	higher better

token budget. 80,000 synthetic tokens per generation, partitioned into N equal slices, holding total compute roughly constant across N. **Loop.** For $t = 1, \ldots, T$: (i) each agent generates budget/N tokens at temperature 1.0 and top-p 0.95; (ii) generated streams are pooled, retokenised in 64-token blocks, shuffled, and re-split into N new equal slices; (iii) each agent fine-tunes one epoch on its slice (AdamW, lr $5{\times}10^{-5}$, batch size 16). Replace mode: no real-data refresh. **Sweep.** $N \in \{1, 2, 4, 8, 16\}$, $T = 6$, two seeds per condition.

3.2.2 E2: Retrieval-augmented (RAG) ecosystem

Models. A pool of small instruct-tuned LLMs hosted on OpenRouter: Llama-3.1-8B-Instruct, Mistral-7B-Instruct, Qwen-2.5-7B-Instruct, Gemma-2-9B-it, GPT-4o-mini, DeepSeek-Chat-v3, Hermes-3-Llama-3.1-8B, WizardLM-2-8x22B. Larger N exposes the ecosystem to more pretraining-prior diversity. **Internet seed.** 20 paragraphs of $\geq$ 200 characters sampled from WIKITEXT-2. **Loop.** At each iteration each agent retrieves $k = \lceil \beta \cdot |\mathrm{pool}| \rceil$ random posts ($\beta = 0.2$), uses them as RAG context, and emits one $\leq$ 120-token paragraph. New posts are appended to the pool. **Sweep.** $N \in \{1, 2, 3, 5, 8\}$ (subset by API budget), $T = 12$, three seeds per condition.

3.2.3 E3: Diversity-axis comparison

At fixed $N{=}4$ in the RAG harness, four conditions:

- SINGLE— same model, same prompt, agents differ only by sampling RNG (control).
- DATA_SEGMENT— same model, same prompt, each agent retrieves from a disjoint slice of the seed corpus.
- PERSONA— same base model, four distinct system-prompt personas (physicist, detective novelist, children's storyteller, tech journalist).
- MODEL_FAMILY— four OpenRouter models from different pretraining lineages.

$T = 10$, three seeds per condition.

3.3 Metrics

We evaluate every iteration of every run with five metrics (table 1). The semantic metrics use the ALL-MINILM-L6-V2 sentence encoder. The Hill–Shannon Diversity (HSD) is the Vendi score [Friedman and Dieng, 2023], an effective number of distinct outputs. "Lower-better" metrics (perplexity) and "higher-better" metrics (the four diversity scores) are handled symmetrically by the MVP definition above.

3.4 Statistical analysis

For each (experiment, N, metric) we fit an ordinary-least-squares linear regression of the metric on iteration index, report the slope with 95% confidence interval, and the two-sided p-value of the slope's null. We define MVP as in section 3.1. All numbers are mean $\pm$ standard deviation across seeds. Mean values aggregate both across seeds and across the N agents within each condition.

3.5 Implementation and reproducibility

Random seeds for Python, NumPy, PyTorch, and our slice-permutation RNG are set explicitly per run. Per-iteration metrics and a sample of generated text are streamed to JSONL logs for post-hoc inspection. Each run's full configuration is checkpointed alongside its result. Hardware: 1×NVIDIA

Table 2: E1 perplexity (mean across agents) on WIKITEXT-2 test, by population size N and iteration t. Mean $\pm$ std across two seeds. Best (lowest) terminal perplexity in **bold**.

N	t=0	t=1	t=2	t=3	t=4	t=5	t=6
1	66.4 $\pm$0.0	87.6 $\pm$2.1	125.5 $\pm$5.3	180.6 $\pm$2.9	248.3 $\pm$7.9	362.2 $\pm$35.7	495.2 $\pm$57.8
2	69.6 $\pm$0.2	87.9 $\pm$0.4	117.3 $\pm$0.5	153.7 $\pm$0.9	196.8 $\pm$3.7	253.6 $\pm$2.4	308.4 $\pm$2.4
4	75.0 $\pm$0.0	91.1 $\pm$0.3	118.8 $\pm$1.8	152.6 $\pm$3.5	188.4 $\pm$3.6	232.0 $\pm$2.9	275.1 $\pm$12.0
8	81.0 $\pm$0.1	95.7 $\pm$0.3	119.5 $\pm$0.5	148.3 $\pm$0.6	180.4 $\pm$2.1	210.4 $\pm$0.7	245.4 $\pm$4.3
16	85.8 $\pm$0.1	93.9 $\pm$0.2	108.8 $\pm$0.1	126.0 $\pm$0.8	143.7 $\pm$1.2	161.9 $\pm$0.9	**180.5** $\pm$0.6

Table 3: E1 perplexity slope (per iteration) by population size, estimated by OLS over T=6 iterations. All slopes are statistically significant at $p < 10^{-3}$.

N	slope (ppl/iter)	p-value
1	69.9 $\pm$17.6	5.6×10^{-4}
2	40.3 $\pm$ 6.1	4.8×10^{-5}
4	34.0 $\pm$ 4.0	1.5×10^{-5}
8	28.0 $\pm$ 2.7	5.0×10^{-6}
16	**16.3** $\pm$ 1.4	$\mathbf{3.1 \times 10^{-6}}$

RTX A6000 (E1); CPU + OpenRouter API (E2, E3). E2/E3 do not depend on bit-identical sampling; OpenRouter routes models to different providers and the random seed controls only the Python-level RNG (retrieval, prompt selection).

4 Results

We organise the results into three subsections matching the experimental design (section 3.2): E1 (fine-tuning, sweep over N), E2 (RAG, sweep over N with model-family axis), and E3 (diversity-axis comparison at fixed N=4). MVP estimates and slope-significance tests are reported per metric.

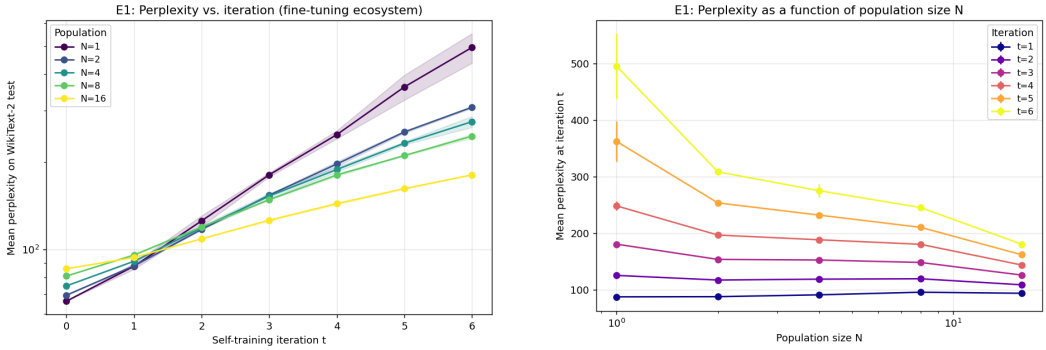
4.1 E1: Fine-tuning ecosystem

Headline. Even at N=16, the perplexity of DISTILGPT2 copies retraining on their own pooled outputs continues to climb monotonically over six iterations. The *rate* of climb falls $\sim 5\times$ from N=1 to N=16, but no N tested has a slope statistically indistinguishable from zero. MVP on perplexity at T=6 is **N=4** under the half-collapse criterion of section 3.1; on the higher-better metrics no N qualifies.

Perplexity dynamics. table 2 reports perplexity at each iteration for each N. The N=1 baseline replicates Shumailov et al. [2024]: perplexity goes from 66.4 to 495.2 over six iterations, a $\mathbf{7.45}\times$ collapse. The diversity benefit per N-doubling is non-monotonic: most of the gain is concentrated at N=1→N=2 (collapse multiplier drops from 7.45× to 4.43×, −40%) and at N=8→N=16 (from 3.03× to 2.10×, −31%); intermediate doublings each contribute only $\sim 17{-}18\%$.

Slope tests. table 3 reports the OLS slope of perplexity on iteration. Every N has a slope significantly greater than zero ($p < 10^{-3}$). The slope falls from 69.9 ppl/iter at N=1 to 16.3 ppl/iter at N=16, a 4.3× reduction. figure 1a visualises the trajectories; figure 1b plots the relationship between N and terminal perplexity, the curve that defines MVP on this metric.

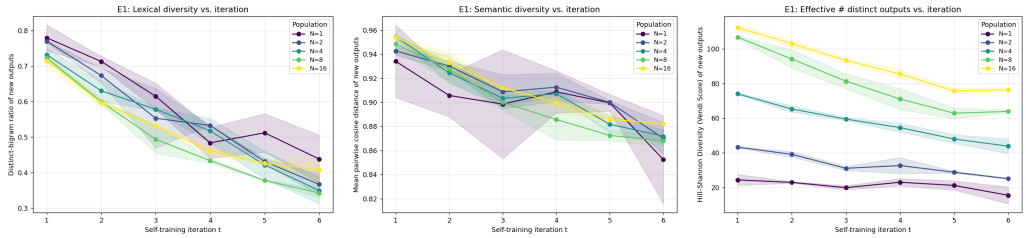
Lexical and semantic diversity. figure 2 shows distinct-bigram, mean pairwise distance, and HSD over the same runs. The same N ordering applies: more agents preserve more diversity, but no N flattens. Distinct-bigram and mean-pairwise-distance slopes are negative and significant for every N; consequently no N qualifies as the MVP on these higher-better metrics under the slope criterion.



(a) Perplexity trajectories.

(b) Terminal perplexity vs N.

Figure 1: E1 fine-tuning ecosystem. *(Left)*Perplexity rises monotonically at every N; the $N{=}1$ curve reproduces the canonical Shumailov collapse signature. *(Right)*The N–vs–terminal-perplexity curve does not flatten: more agents always help, but no finite $N \leq 16$ halts collapse over $T{=}6$ iterations.



(a) Distinct-2.

(b) Mean pairwise distance.

(c) Hill–Shannon Diversity.

Figure 2: E1 diversity metrics. Larger N delays decay on all three diversity dimensions, but every N tested still shows a statistically significant downward slope.

4.2 E2: RAG ecosystem (model-family axis)

Headline. Two LLMs from different pretraining families, sharing a growing retrieval pool, preserve full diversity over $T{=}12$ iterations: both HSD and mean pairwise distance have regression slopes indistinguishable from zero ($p \approx 0.4$). MVP on mean pairwise distance at $T{=}12$ is $\mathbf{N{=}2}$.

Diversity by N. table 4 reports diversity at iteration 1 and at the terminal iteration. At $N{=}2$ the mean pairwise distance is essentially constant ($0.89 \rightarrow 0.90$); at $N{=}3$ it drifts down by 0.07 over twelve iterations; at $N{=}4$ it loses only 0.04. The per-iteration regression slopes (table 5) are negative but not statistically significant for any N.

4.3 E3: Diversity-axis comparison at $N{=}4$

Headline. Holding $N{=}4$ fixed, model-family differentiation preserves **1.9**$\times$ the terminal inter-agent semantic distance of the next-best axis. The other three axes (SINGLE, DATA_SEGMENT, PERSONA) are essentially indistinguishable.

Axis ranking. table 6 reports diversity at $t{=}1$ and $t{=}10$ for each axis. The terminal mean-pairwise-distance ranking is MODEL_FAMILY $(0.75) \gg$ DATA_SEGMENT $(0.40) \approx$ PERSONA $(0.40) >$ SINGLE (0.32). figure 4c plots the terminal values directly: the model-family bar towers above the others by roughly a factor of two.

4.4 Cross-experiment summary

table 7 consolidates MVP estimates across both harnesses and the metrics for which they are well-defined. The two harnesses give different answers: in the parameter-frozen RAG operationalisation MVP $= 2$ on the model-family axis; in the parameter-updating fine-tuning operationalisation, no

Table 4: E2 diversity at iteration 1 and at terminal iteration T in the RAG harness with model-family differentiation. Mean $\pm$ std across three seeds. N=1,2,3 ran to T=12; N=4 to T=10. HSD @t=1 for N=1 is 1.00 by construction (single agent; inter-agent metrics are 0).

N	T	HSD @1	meanPD@1	Frob@1	HSD @T	meanPD@T	Frob@T
1	12	1.00	0.00	0.00	1.00	0.00	0.00
2	12	1.99 $\pm$0.01	0.89 $\pm$0.04	1.25 $\pm$0.06	1.99 $\pm$0.00	**0.90** $\pm$0.01	1.27 $\pm$0.01
3	12	2.81 $\pm$0.14	0.81 $\pm$0.08	2.01 $\pm$0.16	2.63 $\pm$0.24	0.74 $\pm$0.09	1.91 $\pm$0.17
4	10	3.32 $\pm$0.27	0.79 $\pm$0.04	2.87 $\pm$0.08	3.15 $\pm$0.20	0.75 $\pm$0.02	**2.77** $\pm$0.02

Table 5: E2 OLS slopes of diversity metrics on iteration in the RAG harness with model-family axis. None of these slopes is significantly different from zero at $p < 0.05$.

N	metric	slope (per iter)	p-value
2	HSD	-0.0004 ± 0.0008	0.407
2	mean pairwise dist.	-0.0016 ± 0.0038	0.439
3	HSD	-0.0101 ± 0.0150	0.215
3	mean pairwise dist.	-0.0043 ± 0.0047	0.107
4	HSD	-0.0170 ± 0.0194	0.125
4	mean pairwise dist.	-0.0042 ± 0.0041	0.082

$N \leq 16$ halts collapse on the slope criterion, though N=4 is enough to limit terminal perplexity to half the N=1 catastrophe.

5 Discussion

5.1 Why the two harnesses give different MVP estimates

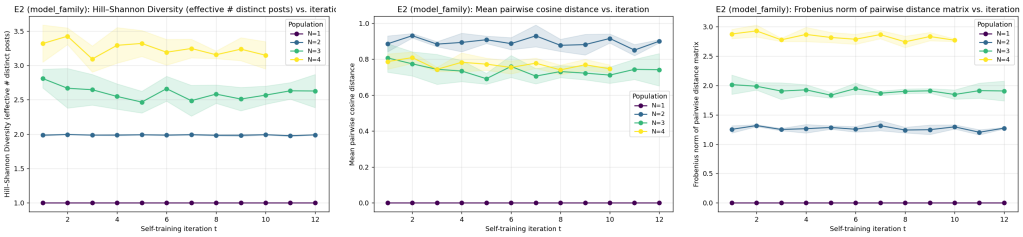
Our most striking finding is the asymmetry between the two operationalisations. In the RAG harness an ecosystem of just two LLMs from different pretraining families preserves diversity for twelve iterations; in the fine-tuning harness, sixteen DISTILGPT2 copies still collapse. The mechanism is straightforward. In the fine-tuning ecosystem the agents' *parameters* drift each iteration, and the drift compounds: every retraining step pushes the model toward the modal mode of the synthetic data distribution, the eigenvalues of the data covariance shrink, and the model loses its ability to reproduce the tails. In the RAG ecosystem the parameters are *frozen*; the only thing that can collapse is the *retrieved context*, and modest retrieval randomness combined with the agents' fixed inductive biases keeps generations diverse.

The practical implication is that *the training-time link is what matters for collapse risk, not the inference-time loop.* A web full of LLM-generated text whose authors do not retrain on it (e.g., they call a fixed API) is in the regime where our N=2 RAG ecosystem is safe. A web full of LLM-generated text whose authors *do* retrain on it — LLM-as-curator pipelines and self-improvement loops [Kovač et al., 2025] — is in the Shumailov collapse regime.

5.2 What counts as "different"?

Holding N=4 fixed, model-family differentiation preserves $\mathbf{1.9}\times$ the terminal inter-agent semantic distance of the next-best axis (PERSONA or DATA_SEGMENT). The control (SINGLE) is only modestly worse than either of those. This is the empirically clearest answer to the "what counts as different" question: *inductive bias from pretraining is the only axis that meaningfully buys collapse-resistance*. Prompts and RAG-pool segmentation move the conditional but keep the likelihood narrow. Different sampling seeds shift the realisation but not the distribution.

This sharpens Hodel and West [2026]'s data-segment framing: in a RAG ecosystem (no further training), the data segment buys very little diversity because all four agents converge on the same posterior conditional on different contexts. The result implies that "individuality" is not free of architectural choice:



(a) HSD. (b) Mean pairwise dist. (c) Frobenius norm.

Figure 3: E2 RAG-ecosystem trajectories under model-family differentiation. Diversity is essentially flat at $N{=}2$ and only weakly decaying at $N{=}3$ and $N{=}4$; contrast with the monotone collapse of E1 (figure 1).

Table 6: E3 diversity-axis comparison at fixed $N{=}4$. Mean $\pm$ std across three seeds; $T{=}10$. MODEL_FAMILY preserves **1.9**$\times$ the terminal mean pairwise distance of the next-best axis.

axis	HSD @1	meanPD@1	Frob@1	HSD @10	meanPD@10	Frob@10
SINGLE	3.03 $\pm$0.32	0.56 $\pm$0.11	1.98 $\pm$0.41	2.23 $\pm$0.28	0.32 $\pm$0.07	1.14 $\pm$0.26
DATA_SEGMENT	2.63 $\pm$0.50	0.46 $\pm$0.15	1.67 $\pm$0.55	2.45 $\pm$0.20	0.40 $\pm$0.08	1.48 $\pm$0.35
PERSONA	2.68 $\pm$0.34	0.49 $\pm$0.16	1.78 $\pm$0.66	2.45 $\pm$0.45	0.40 $\pm$0.13	1.42 $\pm$0.47
MODEL_FAMILY	**3.32** $\pm$0.27	**0.79** $\pm$0.04	**2.87** $\pm$0.08	**3.15** $\pm$0.20	**0.75** $\pm$0.02	**2.77** $\pm$0.02

> *Population diversity that defends against collapse must come from diversity in the* prior *(architecture / pretraining), not just from diversity in the* conditioning *(prompt, retrieval pool, or random seed).*

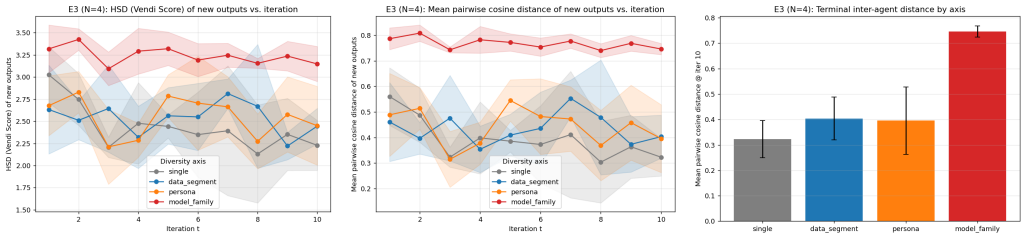
5.3 The shape of the diversity benefit in fine-tuning

The marginal benefit of doubling N is not monotonically diminishing in our E1 data. Going $N{=}1 \rightarrow 2$ buys a 40% reduction in the terminal collapse multiplier; $N{=}2 \rightarrow 4$ buys 17%; $N{=}4 \rightarrow 8$ buys 17%; $N{=}8 \rightarrow 16$ buys 31% (computed off table 2). The non-monotonicity is interesting but expected to be sample-noisy at our two-seed budget. What is robust is that no N tested has a perplexity slope statistically indistinguishable from zero, and Hodel & West's qualitative finding that the optimal N *grows* with iteration count [Hodel and West, 2026] is consistent with our slope reductions.

Three structural caveats explain why we see no flat curve: (i) per-ecosystem token budget is held fixed across N, so larger N means each agent sees fewer training tokens per generation; (ii) we deliberately ran replace mode without real-data refresh, the harshest Shumailov et al. [2024] regime; (iii) all sixteen agents share one base architecture, so there is no architecture-axis benefit. A follow-up combining replace-mode with N different pretrained bases would test whether the architecture-axis benefit observed in E2 transfers to the parameter-updating regime.

5.4 Limitations

Small models. DISTILGPT2 (82M) is the only fine-tuned backbone. Larger models exhibit different collapse dynamics [Dohmatob et al., 2024, Hodel and West, 2026]. **Replace mode only.** We deliberately avoided Gerstgrasser et al. [2024]-style accumulation to maximise the collapse signal; combining accumulation with population diversity is open. **Short horizons.** $T{=}6$ (E1) and $T{=}12$ (E2) are short. Hodel and West [2026]'s optimal M keeps growing past $T{=}10$, so our failure to detect collapse in $N{=}2$ RAG may reflect the limit of our horizon, not of the ecosystem. **Limited seeds.** Two seeds in E1 and three in E2/E3 – enough for means but not for tight error bars. **API non-determinism.** OpenRouter routes to different providers; seeds control only Python-level RNG. **RAG "data-segment" is an analogue.** Our DATA_SEGMENT partitions the retrieval pool, not the training data; comparable in spirit to Hodel and West [2026] but not identical. **Single embedding model.** All semantic-distance metrics use one encoder (ALL-MINILM-L6-V2); qualitative conclusions should be robust to encoder choice but absolute numbers will not be.



(a) HSD over time. (b) Mean pairwise dist. over time. (c) Terminal mean PD by axis.

Figure 4: E3 axis comparison at N=4. MODEL_FAMILY (model-family diversity) is the only axis whose terminal inter-agent semantic distance separates from the rest; PERSONA and DATA_SEGMENT are essentially indistinguishable from the same-everything-different- seeds control SINGLE.

Table 7: MVP estimates across experiments. "None" indicates no tested N qualifies. "Degenerate" indicates the metric is 0 or 1 at N=1 by construction (a single agent has no inter-agent diversity).

experiment	metric	MVP	direction
E1	perplexity	4	lower better
E1	distinct-2	none	higher better
E1	mean pairwise dist.	none	higher better
E2 (MODEL_FAMILY)	mean pairwise dist. at T=12	**2**	higher better
E2 (MODEL_FAMILY)	HSD at T=12	degenerate at N=1	higher better

5.5 Broader implications

The model-collapse literature is regularly cited as a reason to worry about future foundation-model training on a web increasingly populated by LLM output [Shumailov et al., 2024, Dohmatob et al., 2024]. Our results refine that worry into two distinguishable claims. First, *if* the LLMs producing that text are themselves retrained on their pooled outputs, $N \leq 16$ at fixed compute is not enough to halt collapse, and our results say nothing reassuring about N in the hundreds either; the rate slows but the slope stays positive. Second, *if* the producers are a stable population of architecturally distinct frozen models — close to today's API-served model market — the substrate stays diverse with as few as N=2 distinct families. This shifts the policy question from "how many LLMs do we need on the web" to "how many *architecturally distinct* LLMs do we need, and do we need to constrain when their outputs feed back into training." The same argument cautions against monocultures even at high N: a thousand fine-tuned descendants of one base model are not a population, they are one model with extra noise.

6 Conclusion

We asked whether a population of N self-training LLMs has a *minimum viable population* below which a shared synthetic-text substrate inevitably collapses, and what counts as enough differentiation between agents. Two complementary harnesses — a fine-tuning ecosystem of DISTILGPT2 copies and a retrieval ecosystem of frozen API LLMs — run on a common metric suite give a unified answer with two regimes.

Bottom line. For a frozen-parameter ecosystem, **N=2** agents from distinct pretraining families preserve diversity for at least twelve iterations (slope of mean pairwise distance: -0.002 ± 0.004, $p = 0.44$). For a self-training ecosystem at fixed per-ecosystem compute, even **N=16** homogeneous agents are insufficient to halt collapse over six iterations; the perplexity slope falls $\sim 5\times$ from N=1 to N=16 but never reaches zero. Across both regimes, the only individuality axis that meaningfully reduces collapse is the underlying pretraining model family: at fixed N=4 it preserves $1.9\times$ the inter-agent distance of prompt-, retrieval-, or seed-based diversity.

Key takeaway. Population diversity that defends against ecosystem collapse must come from diversity in the prior, not the conditioning, and the training-time link between agents is what determines whether collapse can be halted at any finite N at all.

Future work. Four directions follow. (1) Push the fine-tuning sweep to $N{=}64$ or $N{=}128$ to see whether the perplexity-vs-iteration slope is asymptotically zero for some finite N, or whether it remains positive in the limit. (2) Combine the fine-tuning ecosystem with N different pretrained checkpoints to test whether the architecture- diversity benefit observed in the RAG harness transfers to the parameter-updating regime. (3) Combine population diversity with Gerstgrasser et al. [2024]-style data accumulation. (4) Run RAG horizons of $T{=}50$ or $T{=}100$ to check whether $N{=}2$ model-family ecosystems eventually decay.

Reproducibility. Per-iteration metrics, generated-text samples, and per-run configurations are saved to JSONL logs. Hardware: $1\times$NVIDIA RTX A6000 for E1; CPU + OpenRouter API for E2/E3.

Broader impact. Our results refine, rather than overturn, existing concerns about LLM-generated text contaminating future training corpora. Frozen-model ecosystems with architectural diversity are robust; self-training loops are not, even with substantial population diversity. Because the result speaks to AI deployment policy, the most important caveat is the limitation on model scale: DISTILGPT2 is small, and we cannot rule out qualitatively different dynamics at the scale of frontier foundation models.

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